

For a rxn $A + B \rightarrow C$ Rate = $k[A]^m[B]^n$ where k is the rate constant, m & n are orders of rxn w.r.t A & B respectively. Experimentally determined.

- (a) (i) An equation that mathematically describes how the rate of a chemical reaction depends on the concentrations of the reactants; /AW [1]
 (ii) The power to which the concentrations of reactants are raised in the rate equation; /sum of powers [1]

- (b) (i) H_2O_2 1; [1]
 I^- 1; [1]
 ~~H^+ 1;~~ [1]

- (ii) Rate = $K[H_2O_2][I^-][H^+]$; [1]

- (iii) $mol^{-2}dm^6s^{-1}$; e.c.f. [1]

- (iv) $K = \frac{1.67 \times 10^{-6}}{(0.01)(0.01)(1.0)} = 1.67 \times 10^{-2}$ [1] e.c.f.
 For expt. 3
 $K = 1.75 \times 10^{-2}$

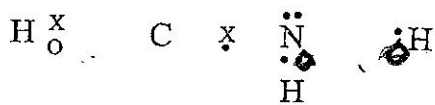
Rate = $(1.67 \times 10^{-2})(0.005)(0.02)(5.0)$
 $Rate = 8.35 \times 10^{-6} mol dm^{-3} s^{-1}$

= $8.35 \times 10^{-6} mol dm^{-3} s^{-1}$; [1]

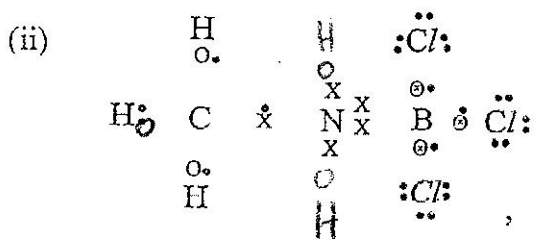
- (c) (i) ~~time for [] of reactant to be halved;~~ [1]
 (ii) ~~be able to control chemical reactions; /AW~~ [1]
 gain insight in the mechanism of a rxn /AW.
 avoid explosions/accidents; /AW [1]
 Help understand processes going on in our bodies / environment.
 maximise yields in chemical reactions; /AW [1]

[Total: 10] II

- 2 (a) (i) $\begin{matrix} H \\ \times O \end{matrix}$



[1]
 © bond should be shown by 2 different e-s must be diff



[1]

- (b) (i) electron pairs place themselves as far apart as possible to minimise repulsion; / *AWA* [1]
 a lone pair exerts a greater repelling effect than a bonded pair; / [1]
 (ii) $\text{LP-LP} > \text{LP-BP} > \text{BP-BP}$ [1]
 trigonal planar; [1]
 120° ; [1]

(c) addition; [1]

(d) lone pairs used in dative bond; / *No lone pair* [1] [2]

[Total: 9] 8

3

(a) (i) A Mg S

B MgCl_2

C Mg(OH)_2

D MgO

epson [1] bath salts ignore use for A+B

[1] medical purposes for mg deficiencies

[1] ingestion remedies / tooth paste

[1] Refractory material / cement / mortar / plaster



for opt use for CO₂ [1] *ignore* state symbol *6 max 5*
wait state symbols

(iii) Mg^{2+} has a greater charge density which results in it having a higher enthalpy of hydration; [1]

This gives a more negative enthalpy of solution; [1]

~~Iodine reacts reversibly at 450°C in the presence of a catalyst;~~ [1]

(b) $\text{Ba(NO}_3)_2$ more stable; [1]

Mg^{2+} has a higher charge density; / *Ba²⁺ has a lower charge density* [1]

and has a greater polarising power; / *it has a smaller polarising power* [1]

[10]

4

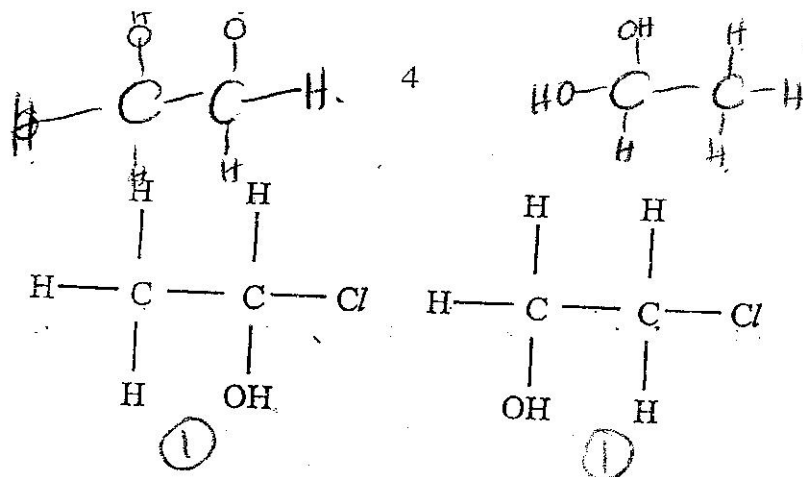
(a) (i) free radical (substitution;) [1]

(ii) when a covalent bond breaks to form two free radicals; / *AWA* [1]

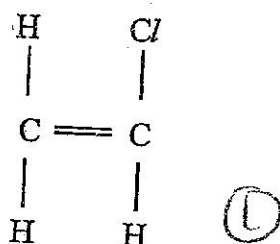
(iii) 1,2 dichloroethane; [1]
 1,1 dichloroethane; [1]

[2]

(b) (i)



(ii)



[1]

[1]

[Total: 7] 7

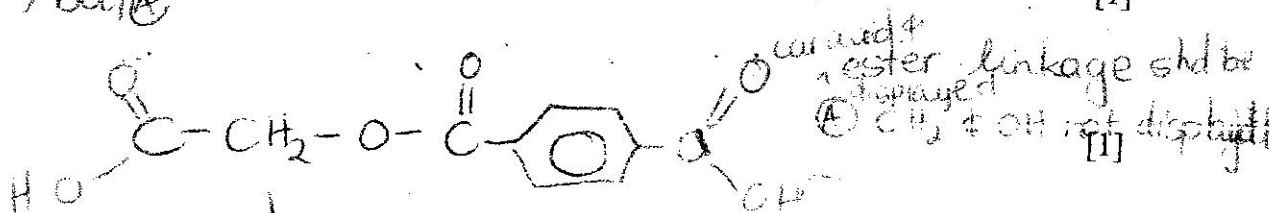
(a) $C_{10}H_{10}O_5$;

[1]

(b) (i) $H^+/K_2Cr_2O_7$ or alkaline $KMnO_4$ or $H^+/KMnO_4$;
heat / reflux;

[1]

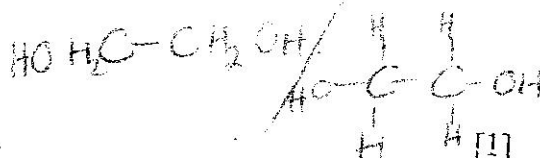
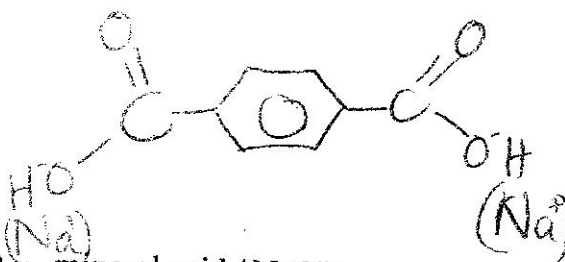
[1]



(ii) mineral acid / NaOH;
reflux; $HCl/H_2SO_4/HNO_3$.

[1]

[1]



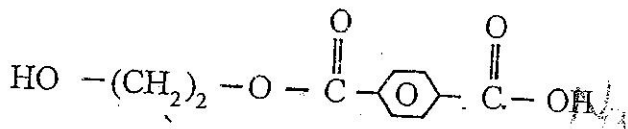
[1]

[1][4]

(iii) mineral acid / NaOH;
room temperature;

[1]

[1]



Wrong reagent don't give a mark for the conditions [Total: 11]